

My Proportion Interactive Notebook

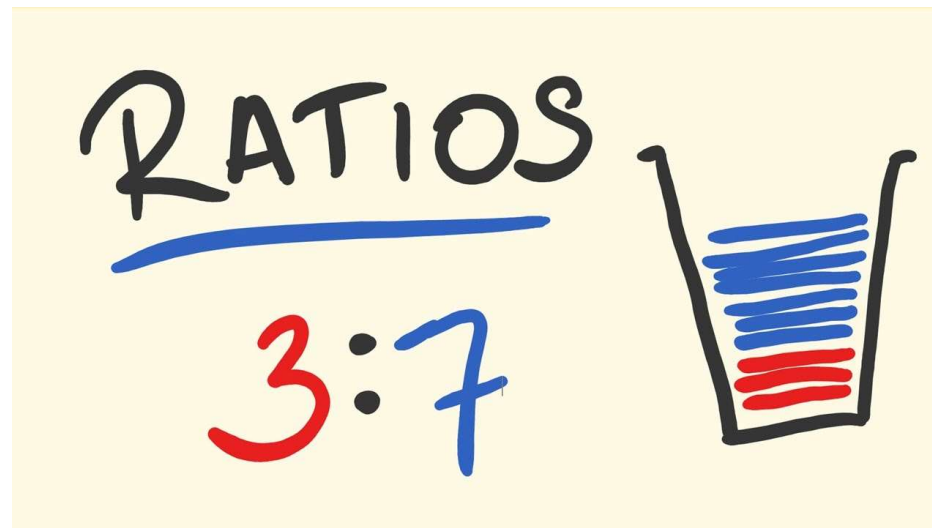
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Name:

Class: 9-9

Term 2 Semester 1



Warm Up

Find the Highest Common Factor (HCF) of the following:

- a) 9 and 12
- b) 16 and 18
- c) 4 and 20
- d) 45 and 54

24
7
30

Date

Direct Proportion

EQ – How can I use direct proportion in my life?

What is direct proportion?

Direct proportion means an increase in one quantity will cause a similar increase in another quantity.

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How can I tell if something is

The direction proportion test: Double one quantity and the other quantity will double.

A pizza scrambled egg recipe uses 6 eggs to serve 4 people.

People (x)	0	4
Eggs (y)	0	6

0 eggs serves 0 people.

Does this look like how we converted tables to formulae???

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0 eggs serves 0 people.

$$y = mx + c$$

Jump of +6 eggs for +4 people

I can use that to find the gradient!

$$\text{Gradient} = \frac{\text{rise}}{\text{run}} = \frac{6}{4} = 1.5$$

So $m = 1.5$

c is the y intercept, and I know it is the value of y when $x = 0$, so in this case $c = 0$

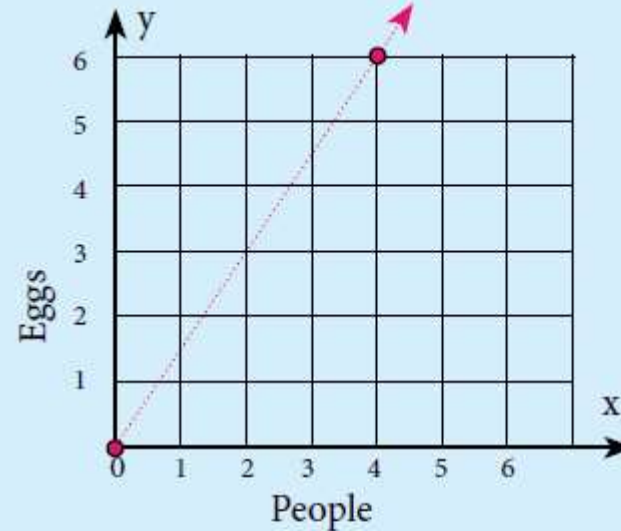
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$$\underline{\text{Eggs} = 1.5 \times \text{people}}$$



Increase of 6 for 4 steps

$$\begin{aligned} \text{so } m &= 6/4 \\ &= 1.5 \end{aligned}$$

$$c = 0 \quad \{\text{cuts y axis at } 0\}$$

Wow, this is all related! #crazy

Work this example together

For each 40 km the rally car
uses 8 litres of fuel.

Distance (x)	0	40
Fuel (y)	0	8

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Change in y = 8 (the rise)

Change in x = 40 (the run)

$$M = \text{gradient} = \frac{\text{rise}}{\text{run}} = \frac{8}{40} = 0.2$$

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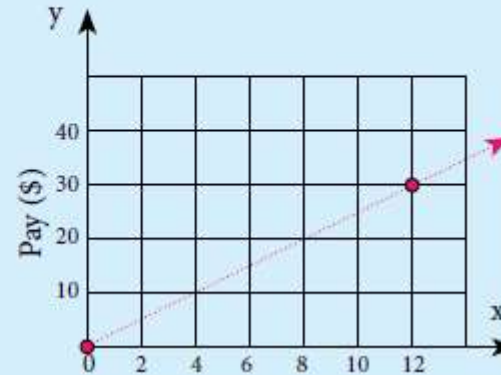
$$\text{Fuel} = 0.2 \times \text{distance}$$

Now, page 78, Ex. 6.6, Q 2-7

Mark was paid \$30 for picking 12 buckets of tomatoes.
How many buckets would Mark need to pick for \$20?

$$y = mx + c \quad \begin{array}{l} \text{Increase of 30 for 12 steps} \\ \text{so } m = 30/12 \\ \quad = 2.5 \end{array}$$
$$c = 0 \quad \{\text{cuts y axis at 0}\}$$
$$\underline{y = 2.5x} \quad \{\text{or } \underline{\text{pay} = 2.5 \times \text{buckets}}\}$$

When pay = \$20



Just like before

$$y = mx + c$$

$$y = 2.5x$$

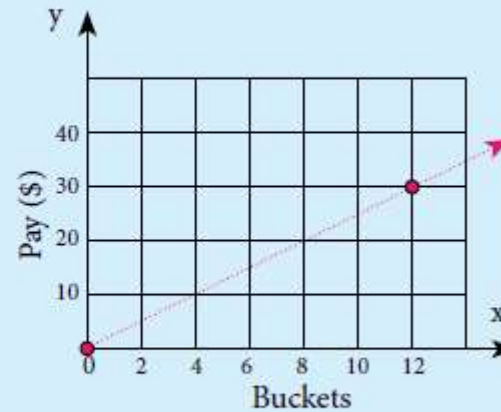
Now we substitute in $y = 20$ to work out the value of x

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$$y = 2.5x \quad \begin{array}{l} c = 0 \quad \{\text{cuts y axis at 0}\} \\ \text{\underline{y = 2.5x}} \quad \{\text{or pay = 2.5} \times \text{buckets}\} \end{array}$$

$$\begin{array}{l} \text{When pay} = \$20 \\ 20 = 2.5 \times \text{buckets} \\ 20 \div 2.5 = \text{buckets} \quad \{\text{inverse of } \times \text{ is } \div\} \\ \underline{8 = \text{buckets}} \end{array}$$



Now lets work on page 78, Ex. 6.6, Q 2-7

Complete your Summary ensuring you explain the process of using direct proportions and how you will apply it in your life.

Notes Review

- 24, 7 ,30
- Date
- Title
- Essential question
- Notes complete
- Notes complete with colour and diagrams
- Questions completed
- Contents page filed in
- Formula page filled in

Homework

- Review your notes and write a summary answering the essential question.
- Ex 6.6, 6.7